E. Multithreading

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given the following concurrent program. There are N processes and the i-th process has the following pseudocode:

```
repeat n_i times
    y_i := y
    y := y_i + 1
end repeat
```

Here y is a shared variable. Everything else is local for the process. All actions on a given row are atomic, i.e. when the process starts executing a row it is never interrupted. Beyond that all interleavings are possible, i.e. every process that has yet work to do can be granted the rights to execute its next row. In the beginning $y = 0$. You will be given an integer W and n_i, for $i = 1, \ldots, N$. Determine if it is possible that after all processes terminate, $y = W$, and if it is possible output an arbitrary schedule that will produce this final value.

Input

In the first line of the input you will be given two space separated integers N ($1 \le N \le 100$) and W ($-10^9 \le W \le 10^9$). In the second line there are N space separated integers n_i ($1 \le n_i \le 1000$).

Output

On the first line of the output write `Yes` if it is possible that at the end $y = W$, or `No` otherwise. If the answer is `No` then there is no second line, but if the answer is `Yes`, then on the second line output a space separated list of integers representing some schedule that leads to the desired result. For more information see note.

Examples

input Copy

```
1 10
11
```

output Copy

```
No
```

input Copy

```
2 3
4 4
```

output Copy

```
Yes
1 1 2 1 2 2 2 2 2 1 2 1 1 1 1 2
```

input Copy

```
3 6
1 2 3
```

output Copy

```
Yes
1 1 2 2 2 2 3 3 3 3 3 3
```

Note

For simplicity, assume that there is no repeat statement in the code of the processes, but the code from the loop is written the correct amount of times. The processes are numbered starting from 1. The list of integers represent which process works on its next instruction at a given step. For example, consider the schedule `1 2 2 1 3`. First process `1` executes its

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 26 (Codeforces format)

Finished

Practice

→ Virtual participation

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Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

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→ Last submissions

Submission	Time	Verdict
372304084	Apr/24/2026 06:45	Accepted
372304002	Apr/24/2026 06:43	Accepted
372303878	Apr/24/2026 06:41	Wrong answer on test 1

first instruction, then process 2 executes its first two instructions, after that process 1 executes its second instruction, and finally process 3 executes its first instruction. The list must consists of exactly $2 \cdot \Sigma_{i=1 \ldots N} n_i$ numbers.

372303794	Apr/24/2026 06:39	Wrong answer on test 3

→ Problem tags

constructive algorithms *2400

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1
- Tutorial #2 (ru)

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